

Optimal-transport flow matching as an alternative to CycleGAN

Unpaired PYTHIA → JEWEL translation of clean calorimeter jet images

October 2026 (CycleGAN: 2-hour results; 8 and 24 hours pending)

seed 0; 20k held-out jets per sample; full log: FLOW_NOTES.md, results: docs/flow/translation

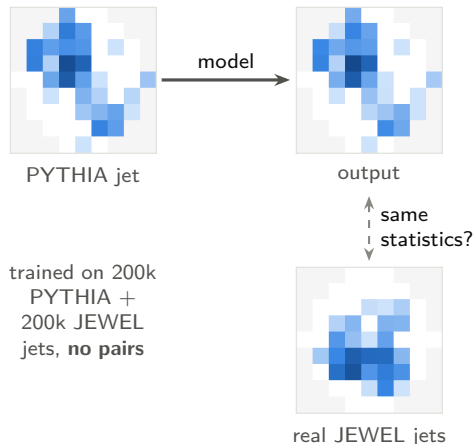
The question

Can optimal-transport flow matching replace a CycleGAN for **unpaired** jet translation: clean PYTHIA (vacuum) $\rightarrow$ JEWEL-like (medium)?

Two things must hold:

1. the outputs look like real JEWEL jets, statistically;
2. each output still depends on its own input jet.

There is no true JEWEL partner for a PYTHIA jet: any unpaired map is one of many with the same distributions. So this tests distributions and dependence, not a per-jet medium modification.



Two ways to learn an unpaired map

— OT-FM [1, 2]

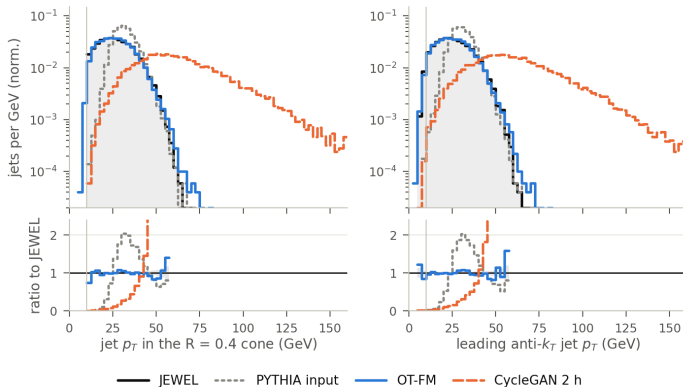
- pair each batch of 256 PYTHIA and 256 JEWEL jets by optimal transport (least change in log energy)
- regress the velocity along straight lines between the pairs: a plain mean-squared-error loss, no adversary
- translate: solve the flow ODE, 128 network calls

— **CycleGAN** [3–5]: the repository's UVCGAN2, which UVCGAN-S builds on

- two generators (PYTHIA $\rightarrow$ JEWEL, JEWEL $\rightarrow$ PYTHIA) and two discriminators
- adversarial, cycle-consistency and identity losses
- translate: one pass of the PYTHIA $\rightarrow$ JEWEL generator

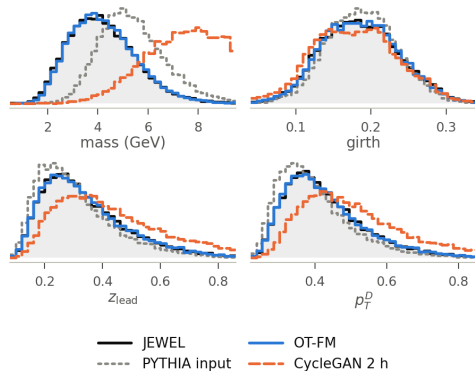
Held equal: the same 200k + 200k training jets (the leading $R = 0.4$ jet, 53 towers on a 16×16 grid), the same log-energy coordinates, the same generator network (the UVCGAN-S ViT-ModNet, 32M parameters), seed 0, and **2 hours on one A6000**. CycleGAN also continues to 8 and 24 hours.

Jet p_T spectra



- **OT-FM** reproduces the JEWEL jet p_T spectrum within statistics (cone and re-found jets), although its inputs start near 20 GeV
- **CycleGAN** at the same 2 hours: p_T about doubled (mean 65 GeV; JEWEL 27, inputs 32); its 1-hour checkpoint lowered it instead ($\times 0.67$): its energy scale had not settled

Jet substructure

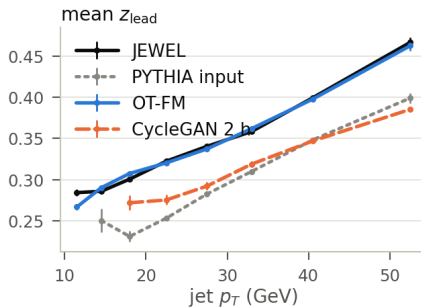
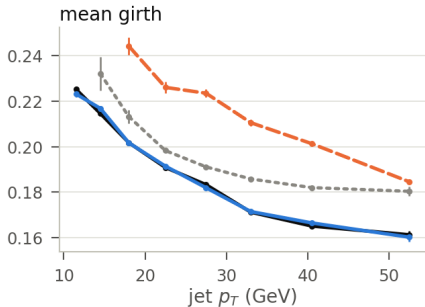


	JEWEL vs JEWEL	PYTHIA input	random JEWEL jet	OT-FM	CycleGAN 2 h
p_T	0.011	0.55	0.019	0.025	3.91
mass	0.014	0.88	0.024	0.021	3.33
girth	0.009	0.10	0.011	0.012	0.14
z_{lead}	0.019	0.23	0.011	0.008	0.58
p_T^D	0.018	0.26	0.010	0.010	0.66
z_g	0.012	0.034	0.005	0.017	0.12
R_g	0.008	0.16	0.010	0.012	0.22

Distance to 20k held-out JEWEL jets ($W1 / \sigma$, lower is better; jets with $p_T \geq 10$ GeV). Bold: as close as a second JEWEL sample, within 3 standard deviations.

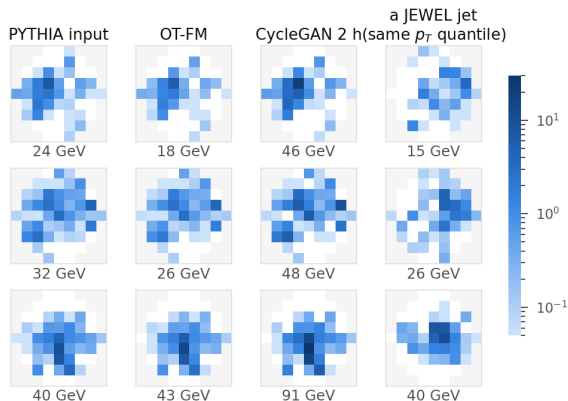
- **OT-FM**: all seven observables as close to JEWEL as a second JEWEL sample
- **CycleGAN** at 2 hours: no observable closer to JEWEL than the PYTHIA input itself; its cores are too hard (leading tower 30 GeV against 9.7)

Jet shapes at fixed p_T



- at the same p_T , JEWEL jets are **narrower** and **harder** (higher z_{lead}) than PYTHIA jets
- **OT-FM** follows both trends in every p_T bin
- **CycleGAN** at 2 hours: at fixed p_T its girth is further from JEWEL than the input's; z_{lead} moves part of the way

Does each output keep its input jet?

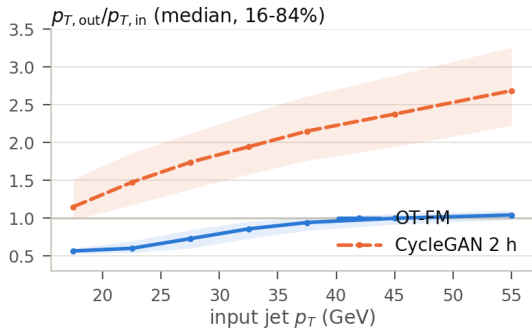


	random JEWEL jet	OT-FM	CycleGAN 2 h
p_T	0.01	0.94	0.85
girth	0.00	0.99	0.99
mass	0.01	0.91	0.85
core fraction	0.00	0.99	0.99
leading tower	0.00	0.92	0.92
own input preferred	50%	100%	100%

Input-output correlation over 20k test jets; own input: share of outputs closer in shape to their own input than to another, random input jet.

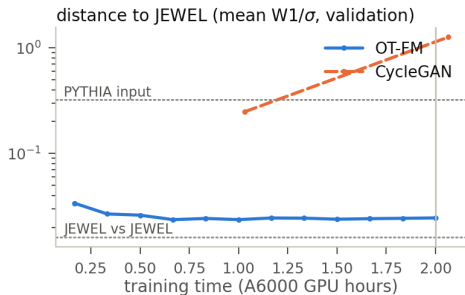
- **OT-FM**: correlations 0.91-0.99; every output is closer to its own input than to another jet
- **CycleGAN** keeps the input too: correlations 0.85-0.99, 99.9% of outputs closest to their own input

Caveat: the p_T change follows the two samples



- the PYTHIA sample starts near 20 GeV (presumably its generator-level jet requirement); JEWEL reaches down to 10 GeV
- **OT-FM** lowers 20-25 GeV jets by 40% and leaves jets above 40 GeV unchanged: a map between two differently selected samples, not a quenching mechanism
- **CycleGAN** at 2 hours raises p_T by a factor growing from 1.2 (15-20 GeV inputs) to 2.6 (above 50 GeV): not a map between the two spectra

Training cost



	GPU h	updates	updates/s	network calls	ms / jet
OT-FM	2.0	123k	17.0	128	6.58
CycleGAN 2 h	2.1	20k	2.7	1	0.05

One A6000; training time without checkpoints; latency at batch 2000.

- **OT-FM**: 123k updates in 2 GPU hours (17 per second), one network pass each
- **CycleGAN**: 2.7 updates per second (six generator passes and two discriminators per update), 20k updates in the same 2 hours; but one network pass per jet, 130 times faster to apply

Summary

- **At the same 2 GPU hours: OT-FM** (no pairs) produces jets statistically indistinguishable from held-out JEWEL (p_T , 7 substructure observables, their p_T dependence, re-found jets), and each output stays tied to its input
- **CycleGAN** (the repository's UVCGAN2, same data and generator network) does not: its p_T scale is off by a factor 2 and no observable moves toward JEWEL; its outputs do keep their inputs
- its training had not settled at 2 hours (the 1-hour checkpoint erred the other way); its 8- and 24-hour checkpoints are being trained, and this deck is rebuilt with them
- **OT-FM's** p_T change reflects how the two samples were selected, not quenching; no per-jet truth exists

References

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